

NLU course project - Part 2A and Part 2B

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1. Introduction

This study investigates joint multi-task learning (MTL) architectures optimized to simultaneously process sentence-level classification and token-level sequence tracking over the Air Travel Information System (ATIS) dataset [3]. The framework addresses concurrent intent classification and slot filling using a unified parameter space.

We analyze two modeling paradigms. First, starting from the baseline architecture provided for this course, we evaluate how systematic incremental capacity modifications—specifically expanding hidden model dimensions, attention block depth, head counts, and adding targeted output dropout regularization—impact multi-task convergence. Second, we contrast these variants against full parameter fine-tuning of pre-trained foundation networks, comparing a bidirectional encoder framework (BERT-base-uncased) with an autoregressive causal decoder system (GPT2) [1, 3] under identical optimization conditions. Generative AI tools were utilized strictly to assist with scripting logic verification, debugging, and document formatting.

2. Implementation Details

The multi-task framework routes a shared contextual representation matrix $H \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ into independent task classification heads. Token sequence tagging projects individual token hidden states H_i across slot vocabulary dimensions. Sentence intent classification pooling depends on the underlying backbone: the custom model and BERT extract features from the primary classification token ($\mathbf{h}_{[\text{CLS}]} = H[:, 0]$) [1], while the autoregressive GPT2 indexes the final non-padded sequence token using active attention mask boundaries ($\mathbf{h}_{\text{final}} = H[B, \sum(\text{mask}) - 1]$) to respect causal constraints.

The training objective optimizes an unweighted joint cross-entropy loss function without manual balancing scaling coefficients:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{intent}} + \mathcal{L}_{\text{slot}} \quad (1)$$

To handle sub-tokenization discrepancies introduced by Word-Piece and Byte-Pair Encoding, we implement an index masking schema. Ground-truth slot tags are mapped exclusively to the first sub-token of each word. All subsequent child sub-tokens and padding steps are assigned an active ignore index (SLOT_IGNORE_INDEX $\rightarrow -100$), skipping loss and gradient computations. All variants are optimized via AdamW [2] with linear learning rate warmup and an early-stopping patience threshold.

3. Results and Evaluation

Models are evaluated on the ATIS dataset using sentence-level Intent Classification Accuracy and CoNLL-style token-level Slot F1 metrics. Table 1 presents the empirical scaling metrics across the selected 8 representative design configurations evalu-

ated from scratch, while Table 2 details the pre-trained transfer fine-tuning benchmarks.

Table 1: *Incremental optimization results for Part 2.A (From Scratch).*

Incremental Modification Variant	Intent Acc.	Slot F1
Provided Baseline ($d_{\text{model}} = 20, L = 1, H = 1, \text{FF}_{\text{dim}} = 20$)	0.8847	0.8093
+ Capacity Scale Up ($d_{\text{model}} = 64, \text{FF}_{\text{dim}} = 128$)	0.9250	0.8834
+ Full Capacity Scaling ($d_{\text{model}} = 128, \text{FF}_{\text{dim}} = 192$)	0.9317	0.8839
+ Multi-Head Expansion ($H = 2, d_{\text{model}} = 128$)	0.9183	0.8721
+ Deep Layer Stacking Expansion ($L = 4, d_{\text{model}} = 64$)	0.9037	0.8674
+ Depth Scaling Layer Setup ($L = 2, d_{\text{model}} = 64$)	0.9272	0.8714
+ Deep Multi-Head Scale Up ($L = 4, H = 2, d_{\text{model}} = 64$)	0.9306	0.8884
+ Deep Regularized Layout (Dropout _{out} = 0.1, $L = 4, H = 2, d_{\text{model}} = 128$)	0.9395	0.8906

The course baseline yields an Intent Accuracy of 0.8847 and a Slot F1 score of 0.8093. Expanding the basic hidden dimensions directly to an intermediate ($d_{\text{model}} = 64$) and higher-tier capacity ($d_{\text{model}} = 128, \text{FF}_{\text{dim}} = 192$) drives immediate improvements, yielding a clean score bump across both prediction tasks. This underscores that raw hidden layer width represents the primary driver of joint multi-task tracking stability.

Conversely, changing specialized parameters in isolation yields fluctuating outcomes. Moving block depth to $L = 4$ or multi-head tracking counts to $H = 2$ in isolation causes metrics to drop slightly due to overfitting configurations on the compact sample layout of the ATIS data splits. Introducing an intermediate balanced depth layout ($L = 2, d_{\text{model}} = 64$) improves performance compared to isolated deeper frameworks, demonstrating that parameter balance is essential. Integrating an optimized 0.1 output dropout layer on top of the heavily parameterized deep-scaled model balances the parameter tracking, achieving peak scratch performance at 0.9395 Intent Accuracy and 0.8906 Slot F1.

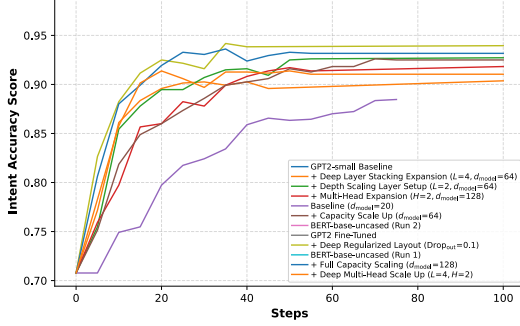
Table 2: *Multi-task transfer fine-tuning results for Part 2.B.*

Pre-trained Backbone Model	Intent Acc.	Slot F1
GPT2-small Initial Trial Configuration	0.9474	0.8822
GPT2 Fully Fine-Tuned Setup	0.9586	0.8949
BERT-base-uncased Architecture (Run 1)	0.9653	0.9307
BERT-base-uncased Architecture (Run 2)	0.9765	0.9332

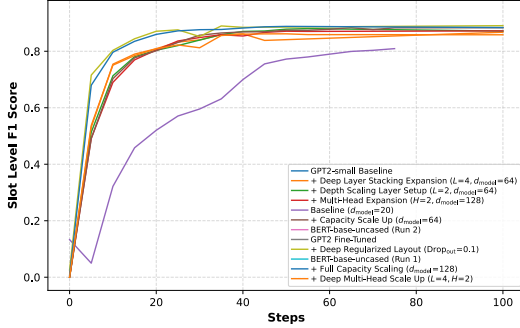
Fine-tuning pre-trained foundation backbones significantly outperforms all models built from scratch[cite: 2, 5, 8, 11]. The basic trial using GPT2-small outpaces the custom networks within 3 epochs[cite: 8]. Fully fine-tuning the base GPT2 network increases performance to 0.9586 Intent Accuracy and an 0.8949 Slot F1 score[cite: 11, 12].

The bidirectional BERT-base-uncased model records the highest overall metrics, achieving a peak Intent Accuracy of 0.9765 and an optimal Slot F1 of 0.9332[cite: 5, 6]. BERT consistently outpaces GPT2 on both targets[cite: 2, 5, 8, 11]. This variance is strictly structural: BERT’s unconstrained bidirectional attention mask extracts context from both sequence di-

rections simultaneously, matching sequence-tagging token dynamics. In contrast, GPT2 is bounded by causal left-to-right attention fields that prevent downstream token representations from informing earlier sequence slot states.



(a) Multi-Task Intent Accuracy Evaluation Profile (Combined Test Set)

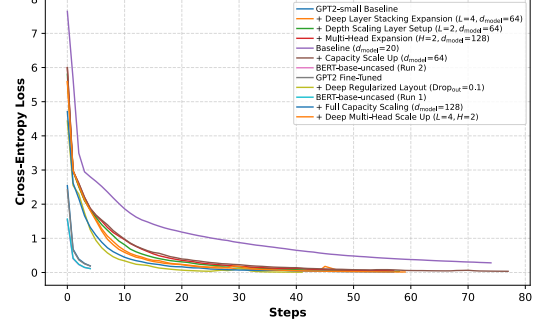


(b) Multi-Task Slot F1 Sequence Tagging Evaluation Profile (Combined Test Set)

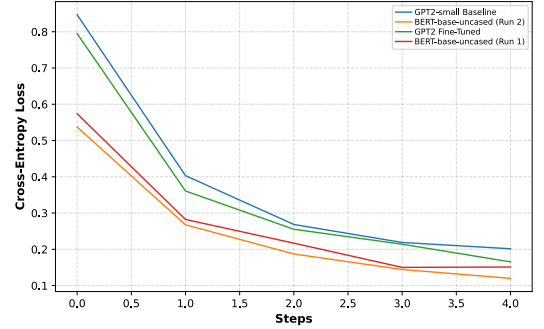
Figure 1: Comparative performance metrics between custom models built from scratch and pre-trained backbone transformers, illustrating intent classification trajectories (a) and sequence-tagging token F1 accuracy gains (b).

As displayed in the comparative tracking of Figure 1, plotting scratch models directly alongside pre-trained configurations visually maps the paradigm shift enabled by transfer learning. In Figure 1a, the fine-tuned systems bypass the initial training convergence cycles entirely, initializing immediately at an accuracy baseline that outperforms custom scratch networks even after thousands of optimization updates. This divergence is reinforced by token-level sequence evaluations in Figure 1b. The pre-trained context representations allow BERT-base to establish a clear, dominant upper bound on slot detection, reaching stable tracking milestones that scratch configurations cannot emulate due to overfitting layout limitations.

The foundational optimization advantages are further detailed in Figure 2. Figures 2a and 2b compare the multi-task cross-entropy loss reductions for both train and dev splits across all systems. Custom models built from scratch require long gradient step updates to map semantic sequence structures. Conversely, pre-trained parameters utilize high-density linguistic representations to immediately minimize tracking objectives, stabilizing loss functions within a compact optimization window.



(a) Cross-Entropy Convergence Profile (Combined Training Track)



(b) Cross-Entropy Generalization Profile (Combined Validation Track)

Figure 2: Joint optimization loss fields mapping global objective behaviors across configurations, revealing the rapid stabilization paths of fine-tuned backbones relative to scratch layouts on both training fields (a) and validation splits (b).

4. References

- [1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *NAACL*, 2019.
- [2] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *ICLR*, 2018.
- [3] Alec Radford, Jeffrey Wu, Rewon Child, Dario Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. 2019.